# Is Exchangeability better than I.I.D. to handle Data Distribution Shifts while Pooling Data for Data-scarce Medical image segmentation?

MIDL Taipei 2026

University at Buffalo

Accepted at MIDL 2026

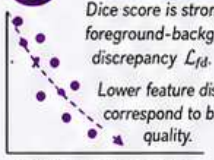
Ayush Roy[1], Samin Enam[1], Jun Xia[1], Won Hwa Kim[2], Vishnu Suresh Lokhande[1]

[1] University at Buffalo (SUNY) [2] Pohang University of Science and Technology (POSTECH)

## 1) PROBLEM STATEMENT

- Medical image segmentation is often data-scarce.
- Pooling data from multiple sources or incrementally adding new data can improve performance, but distribution shifts can also hurt performance.
- This phenomenon is the **"Data Addition Dilemma"**: more data is not always better when the new data distribution is mismatched.
- In multi-source medical imaging, the i.i.d. assumption is often unrealistic; exchangeability is a more practical assumption.

### Key Observation

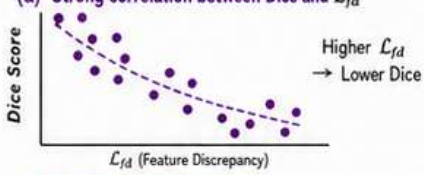



*Dice score is strongly correlated with foreground-background feature discrepancy $\mathcal{L}_{fd}$.*

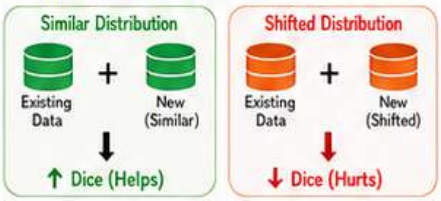
*Lower feature discrepancy tends to correspond to better segmentation quality.*

### Illustrative Motivation

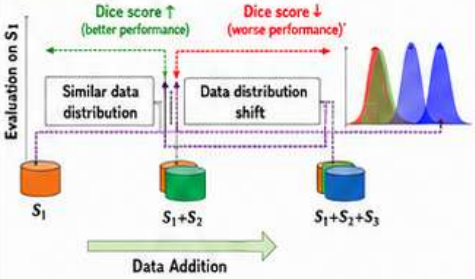
(a) Strong correlation between Dice and $\mathcal{L}_{fd}$




(b) Data Addition Dilemma

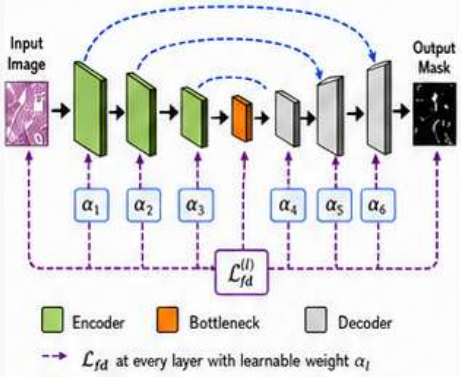



(c) Impact of Distribution Shift on Model Performance




## 2) METHOD

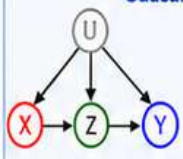
### Overall Framework




### Key Ideas

- Apply Feature Discrepancy Loss $\mathcal{L}_{fd}$ at every layer to improve foreground-background separation.
- Each layer $l$ has a learnable weight $\alpha_l$, warm-started from 0 to stabilize training.
- For data addition under assumed exchangeability, use an exchangeable variant $\mathcal{L}_{fd}^{exch}$ to compare base and novel datasets and mitigate cross-dataset shifts.

### Causal Intuition




X: Input image

Z: Layer feature (mediator)

Y: Segmentation label

U: Unobserved confounders

Improving the mediator Z by reducing $\mathcal{L}_{fd}$ leads to more robust segmentation $Y$.

### Loss Formulation

$$\mathcal{L}_{seg} = \text{Dice} + \text{BCE}$$

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \sum_{l=1}^{L} \alpha_l \, \mathcal{L}_{fd}^{(l)}$$

$$\mathcal{L}_{total}^{exch} = \mathcal{L}_{seg} + \sum_{l=1}^{L} \alpha_l \left( \mathcal{L}_{fd}^{(l)} + \mathcal{L}_{fd,exch}^{(l)} \right)$$

### ★ Contributions

1. *A foreground-background feature discrepancy loss for medical image segmentation.*
2. *An exchangeability-based variant for data addition under distribution shift.*
3. *Theoretical motivation linking improved Dice to lower feature discrepancy.*
4. *Strong experiments on histopathology and ultrasound, including a new US-TNBC dataset.*

## 3) RESULTS

### Summary

Evaluated on 5 datasets: TNBC, MoNuSeg, AD256, UDIAT, and US-TNBC.

Tested across 3 architectures: AttnUNet, CMUNet, and NuSegNet.

$\mathcal{L}_{fd}$ consistently improves Dice and IoU, especially on difficult / worst-off samples.

### Selected Quantitative Results (All Samples)

| Model | Dataset | Dice (%) | | | IoU (%) | | |
|---|---|---|---|---|---|---|---|
| | | w/o $\mathcal{L}_{fd}$ | w/ $\mathcal{L}_{fd}$ | Δ Dice | w/o $\mathcal{L}_{fd}$ | w/ $\mathcal{L}_{fd}$ | Δ IoU |
| AttnUNet | MoNuSeg | 75.92 | 77.97 | +2.05 | 61.28 | 62.87 | +1.59 |
| CMUNet | UDIAT | 81.85 | 84.22 | +2.37 | 69.87 | 73.02 | +3.15 |
| NuSegNet | AD256 | 51.15 | 56.57 | +5.42 | 36.17 | 40.61 | +4.44 |
| CMUNet | US-TNBC | 49.59 | 50.22 | +0.63 | 34.53 | 36.52 | +1.99 |

### Data Addition Dilemma (Figure 3)

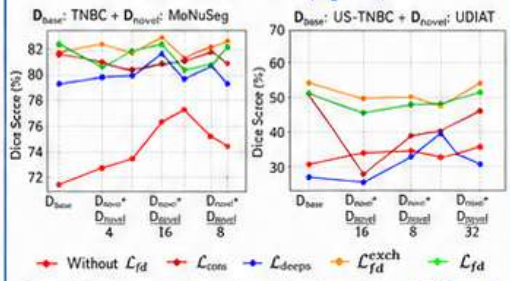



**Figure 3.** Under sequential data addition, the exchangeable loss $\mathcal{L}_{fd}^{exch}$ (orange) best mitigates distribution shift and outperforms the baselines in both histopathology and ultrasound settings.

### Consistent Gains Across Settings

- *Worst-off samples improve noticeably after applying $\mathcal{L}_{fd}$.*
- *Boundaries become sharper and unnecessary activations are reduced.*
- *The exchangeable variant helps mitigate distribution shifts during data addition.*

### Takeaway

*Exchangeability offers a more realistic assumption than i.i.d. for multi-source medical imaging. By penalizing foreground-background feature discrepancy across layers, the proposed method improves robustness and segmentation quality under data scarcity and distribution shift.*

## Conclusion

*We introduce a feature discrepancy penalty that improves medical image segmentation on both histopathology and ultrasound images. The method is particularly effective under the Data Addition Dilemma, where more data can otherwise degrade performance under distribution shift. The exchangeable variant further stabilizes learning when adding new data from a shifted source.*

**Code available on GitHub**

https://github.com/AyushRoy2001/Exchangeable-feature-disentanglement